

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

EE2002-1 – INSTRUMENTATION AND MEASUREMENTS
(E&E)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. A thermometer reads 95.45°C and the static correction given in the correction curve is -0.08°C . The true value of the temperature is _____
 A) 95.37°C B) 95.53°C
 C) 95.29°C D) 95.61°C
2. _____ is defined as variation of scale reading and is random in nature
 A) Reproducibility B) Span drift
 C) Repeatability D) Zero drift
3. Murry loop test is used for the detection of _____
 A) Ground fault B) Short circuit fault
 C) Cable fault D) Both ground and short circuit fault
4. In a Kelvin's Double Bridge two set of readings are taken when measuring low resistance, one with the current in one direction and other with direction of current reversed. This eliminates _____
 A) Effect of thermo-electric emfs B) Effect of contact resistance
 C) Effect of resistance of leads D) Effect of sliding leads
5. Maxwells inductance-capacitance bridge is used for measurement of inductance of _____
 A) Low Q coils B) High Q coils
 C) Medium Q coils D) Low and medium Q coils
6. A 1mA meter movement with an internal resistance of 100Ω is to be converted into a $0-100\text{ mA}$ ammeter, the shunt resistance required is _____
 A) 0.98Ω B) 1.02Ω
 C) 0.99Ω D) 1.01Ω
7. Light load adjustment for induction type energy meters are usually done at _____
 A) 10% of full load current B) 5% of full load current
 C) 50% of full load current D) 1% of full load current
8. A Schering bridge is used for the measurement of _____
 A) Inductance B) Resistance
 C) Capacitance D) Frequency
9. A transducer converts _____
 A) Mechanical energy to electrical energy B) Electrical energy to mechanical energy
 C) One form of energy into another form of energy D) Mechanical displacement to electrical form
10. _____ is considered as the most suitable element in high resistance thermometers
 A) Platinum B) Nickel
 C) Copper D) Gold
11. Capacitance transducers are normally used for _____
 A) Static measurements B) Transient measurements
 C) Dynamic measurements D) Both static and dynamic measurements
12. A Hall effect sensor can be used for the measurement of _____
 A) Proximity B) Position
 C) Speed D) All of these

13. The controlling torque in single phase power factor meter is provided by _____
 A) Spring control B) Gravity control
 C) Suspension control D) None of these
14. A phase sequence indicator rotates clockwise for phase sequence RYB. If the phase sequence is changed to BRY it will rotate in _____ direction
 A) Anticlockwise B) clockwise
 C) Clockwise or anticlockwise D) none of these
15. _____ is directly measured in LVDT
 A) Displacement B) Force
 C) Temperature D) Pressure
16. In optical recording _____ signal is converted to variations of light intensity falling onto the film for recording on photographic film
 A) Light B) Sound
 C) Audio D) Video
17. The switching time of LEDs is in the order of
 A) 1s B) 1ms
 C) 1 μ s D) 1ns
18. A photovoltaic cell has _____ internal resistance
 A) High B) Medium
 C) Low D) Infinite
19. A successive approximation ADC with Vcc of 5V has a resolution of 20mV. What will be the digital output for an analog input of 2.17V?
 A) 01101100 B) 01101101
 C) 01101011 D) 01110101
20. A _____ type of DVM employs voltage comparison technique
 A) Ramp B) Potentiometric
 C) Integrating D) Successive approximation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) List the various static characteristics of measuring instruments. | 04 | | | |
| b) With a neat diagram derive the balancing equation for Anderson's Bridge. | 05 | L*1 | 01 | 1,2 |
| c) A moving coil instrument gives a full-scale deflection of 10mA when the potential difference across its terminals is 100mV. Calculate (i) the shunt resistance for a full-scale deflection corresponding to 100A, (ii) the series resistance for full scale reading with 1000V. Calculate the power dissipation in each case. | 07 | L3 | 02 | 1,2 |
| 2. a) Explain the operation of shunts used to extend the range of an ammeter. | 04 | L2 | 02 | 1,2 |
| b) Discuss the various errors and adjustments in a single phase energy meter. | 05 | L2 | 02 | 1,2 |
| c) A Maxwells capacitance bridge is used to measure unknown inductance in comparison with capacitance. The values at balance are:
Arm AD non inductive resistance (R_2) of 400 Ω
Arm BC non inductive resistance (R_3) of 600 Ω
Arm CD is parallel combination of capacitance (C_4) of 0.5 μ F and non-inductive resistance (R_4) of 1000 Ω
Calculate the values of resistance (R_1) and inductance(L_1) of the coil connected across the arm AB. Also calculate the Q factor of the coil if frequency is 1000Hz.
Also draw its phasor diagram at balance. | 07 | L3 | 03 | 1,2 |
| 3. a) Discuss briefly the measurement of earth resistance with fall of potential method | 04 | L2 | 01 | 1,2 |
| b) In a Wheatstone bridge three out of four resistors have a value of 1K Ω each, and the fourth resistor equals 1010 Ω . If the battery voltage is 100V, what is the approximate value of the open circuit voltage ? If the output of the bridge is connected to a 4k Ω resistor calculate the current flowing through the resistor. | 05 | L3 | 01 | 1,2 |

- c) With a neat block diagram, explain the operation of digital energy meter. Also list its advantages.

07 L2 02 1,2

Unit – II

4. a) List the classification of transducers. 05 L1 03 1,2
 b) With a neat circuit diagram, discuss the operation of Weston frequency meter. 05 L2 03 1,2
 c) Describe the basic components of magnetic tape recorder used for instrumentation applications using direct recording techniques. Also list its advantages. 06 L2 04 1,2
5. a) Discuss the construction and working of rotating type phase sequence indicator. 05 L2 03 1,2
 b) Explain the construction and working of fiber optic sensor. 05 L2 03 1,2
 c) Describe the temperature measurement method using a thermistor. 06 L2 04 1,2
6. a) Distinguish between RZ and NRZ techniques of digital tape recording. 05 L1 04 1,2
 b) Describe the features and types of rotary encoders. 05 L2 04 1,2
 c) Discuss the construction of strain gauge and derive an expression for gauge factor. 06 L3 03 1,2

Unit – III

7. a) Discuss the principle of measurement of phase using Lissajous patterns. 05 L2 05 1,2
 b) With a neat diagram explain the operation of binary weighted digital to analog converter. 05 L2 05 1,2
 c) Describe the circuit diagram and operation of true rms responding voltmeter using thermocouples. 06 L2 05 1,2
8. a) List the advantages and disadvantages of LCD display. 05 L1 05 1,2
 b) Describe the functioning of ramp type digital voltmeter. 05 L2 05 1,2
 c) With a neat block diagram, explain the operation of digital storage oscilloscope. 06 L2 05 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
